Technical Documentation: Marketing Capstone Project

1. Project Overview

Objective:

A restaurant consolidator aims to revamp its B2C portal using intelligent automation to identify “star restaurants” and generate personalised recommendations.

This project analyses restaurant-level data to understand customer behaviour, performance drivers, and operational signals.

Business Goals:

- Identify high-performing (star) restaurants

- Understand factors influencing ratings and customer engagement

- Support recommendation logic for B2C portal revamp

- Enable data-driven decision-making via dashboards

2. Datasets Used

Dataset 1: Restaurant Data

- 19 attributes

- Key columns: Restaurant ID, Restaurant Name, City, Country Code, Cuisines, Cost for Two, Aggregate Rating, Votes, Online Delivery, Table Booking

Dataset 2: Country-Code

- 2 attributes: Country Code, Country Name

- Used for geographical enrichment

3. Python: Data Import & Inspection

Step 1: Import Required Libraries

- pandas, numpy, matplotlib, seaborn

Step 2: Load Datasets

- Loaded restaurant dataset and country-code dataset into pandas DataFrames

Step 3: Preliminary Inspection

- Used .head(), .info(), .shape() to understand structure

- Verified column data types

- Checked for inconsistent data formats

Screenshot : Please find detailed code and charts in MarketingCapstone.ipynb file.

```
data.shape
data.columns
```

```
Index(['Restaurant ID', 'Restaurant Name', 'Country Code', 'City', 'Address',
       'Locality', 'Locality Verbose', 'Longitude', 'Latitude', 'Cuisines',
       'Average Cost for two', 'Currency', 'Has Table booking',
       'Has Online delivery', 'Price range', 'Aggregate rating',
       'Rating color', 'Rating text', 'Votes'],
      dtype='object')
```

```
# str of data

data.shape

data.info()

data.isnull().sum()

print(data.duplicated().sum())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9551 entries, 0 to 9550
Data columns (total 19 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   Restaurant ID         9551 non-null   int64
 1   Restaurant Name       9550 non-null   object
 2   Country Code          9551 non-null   int64
```

Step 4: Missing Values Analysis

- Used .isnull().sum() to identify missing values

- Evaluated impact of missing data on analysis

Please find detailed code and charts in MarketingCapstone.ipynb file.



Count missing values

```
data.isnull().sum()
```

```
Restaurant ID           0
Restaurant Name         1
Country Code            0
City                    0
Address                 0
Locality                0
Locality Verbose        0
Longitude               0
Latitude                0
Cuisines                9
Average Cost for two    0
Currency                0
Has Table booking       0
Has Online delivery     0
Price range             0
Aggregate rating        0
Rating color            0
Rating text             0
Votes                   0
dtype: int64
```

```
data.isnull().sum()/len(data)*100
```

```
Restaurant ID           0.000000
Restaurant Name         0.010470
Country Code            0.000000
City                    0.000000
```

Step 5: Duplicate Detection

- Identified duplicate records using Restaurant ID

- Validated duplicates across key descriptive columns

- Removed duplicates using drop_duplicates()

Please find detailed code and charts in MarketingCapstone.ipynb file.



```
data.duplicated().sum()
```

```
np.int64(0)
```

```
data[data.duplicated()]
```

Restaurant ID	Restaurant Name	Country Code	City	Address	Locality	Locality Verbose	L

```
data=data.drop_duplicates()

print(data.duplicated().sum())
```

```
0
```

Outcome:

- Clean and reliable dataset prepared for analysis

4. Python: Data Preparation & Integration

Step 1: Dataset Merge

- Joined Restaurant Data with Country-Code on Country Code

- Join Type: Left Join

Step 2: Feature Preparation

- Converted Online Delivery and Table Booking flags to categorical/boolean values

- Ensured ratings, votes, and cost columns were numeric

- Created cuisine count feature by splitting cuisine strings

Step 3: Data Validation

- Verified row counts post-merge

- Ensured no unintended data loss

Please find detailed code and charts in MarketingCapstone.ipynb file.

Merge Country Name

```
data_merged=pd.merge(
    data,country_code,
    on='Country Code',
    how='left')
```

```
data_merged.head()
```

	Restaurant ID	Restaurant Name	Country Code	City	Address	Locality	I V
0	7402935	Skye	94	Jakarta	Menara BCA, Lantai 56, Jl. MH. Thamrin, Thamri...	Grand Indonesia Mall, Thamrin	In T
1	7410290	Satoo - Hotel Shangri-La	94	Jakarta	Hotel Shangri-La, Jl. Jend. Sudirman	Hotel Shangri-La, Sudirman	Sha Su
2	7420899	Sushi Masa	94	Jakarta	Jl. Tuna Raya No. 5, Penjaringan	Penjaringan	Penj
					Jl. Suryo		

5. Python: Exploratory Data Analysis (EDA)

Geographical Analysis:

- Grouped restaurants by Country and City

- Identified cities with maximum and minimum restaurant counts

Please find detailed code and charts in MarketingCapstone.ipynb file.

City with max restaurants

```
city_dist.idxmax(),city_dist.max()
```

```
('New Delhi', 5473)
```

City with min restaurants

```
city_dist.idxmin(),city_dist.min()
```

```
('Bandung', 1)
```

```
city_dist.head(10).plot(
    kind='bar',
    figsize=(10,4),
    title='Top 10 cities by number of restaurants'
)
plt.ylabel('Number of restaurants')
plt.show()
```

Franchise Analysis:

- Grouped by Restaurant Name

- Calculated unique country presence for each franchise

Please find detailed code and charts in MarketingCapstone.ipynb file.

Group by Restaurant Name and count countries

```
franchise_presence = (
    data_merged.groupby('Restaurant Name')['Country'].nunique().sort_values(ascending=False)
)
```

Top franchise by national presence

```
franchise_presence.head(10)
```

```
Restaurant Name
Hard Rock Cafe         3
Indian Coffee House    2
Ichiban                2
Gazebo                 2
Jamie's Italian        2
Din Tai Fung           2
Barbeque Nation        2
Gymkhana               2
Chili's Grill & Bar    2
Applebee's             2
Name: Country, dtype: int64
```

Franchise with most national presence

```
top_franchise = franchise_presence.idxmax()
top_franchise_countries = franchise_presence.max()
top_franchise,top_franchise_countries
```

```
('Hard Rock Cafe', 3)
```

Table Booking & Delivery Analysis:

- Calculated count and percentage of restaurants offering:

  - Online Delivery

  - Table Booking

Please find detailed code and charts in MarketingCapstone.ipynb file.

Ratio : Table booking Vs No Table booking

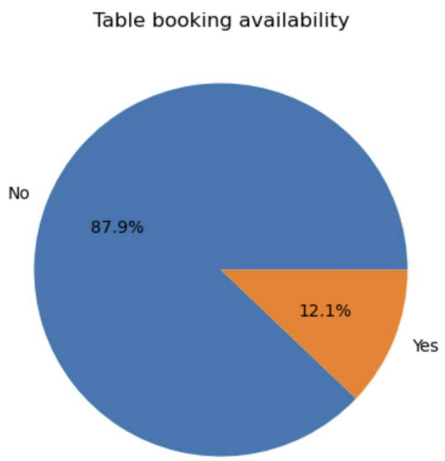
```
table_booking_counts = data_merged['Has Table booking'].value_counts()
table_booking_counts
```

```
Has Table booking
No     8393
Yes    1158
Name: count, dtype: int64
```

```
table_booking_ratio = table_booking_counts/table_booking_counts.sum()
table_booking_ratio
```

```
Has Table booking
No     0.878756
Yes    0.121244
Name: count, dtype: float64
```

```
table_booking_counts.plot(
    kind='pie',
    autopct='%1.1f%%',
    figsize=(5,5),
    title = 'Table booking availability'
)
plt.ylabel('')
plt.show()
```



Votes Analysis:

- Compared average and total votes for delivery vs non-delivery restaurants

Please find detailed code and charts in MarketingCapstone.ipynb file.

Aggregate Votes By Delivery Options

```
votes_by_delivery = (
    data_merged.groupby('Has Online delivery')['Votes'].sum()
)
votes_by_delivery
```

```
Has Online delivery
No     980731
Yes    517914
Name: Votes, dtype: int64
```

Difference in Votes

```
vote_difference = votes_by_delivery.max() - votes_by_delivery.min()
vote_difference
```

```
462817
```

Cuisine Analysis:

- Identified top 10 cuisines across cities

- Calculated max and min cuisines per restaurant

- Determined most common cuisine per city

Please find detailed code and charts in MarketingCapstone.ipynb file.

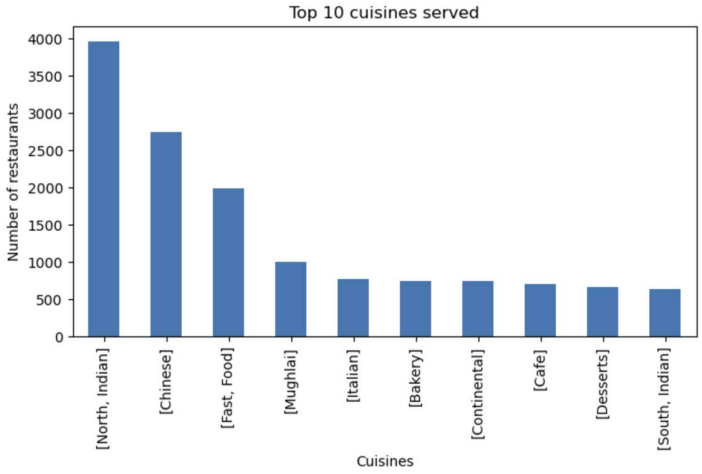
Top 10 cuisines served across cities

```
# split cuisines into individual value

cuisine_series = (
    data_merged['Cuisines'].dropna().str.split(',').explode().str.split()
)
```

```
# top 10 cuisines overall
top_10_cuisines = cuisine_series.value_counts().head(10)
top_10_cuisines
```

```
Cuisines
[North, Indian]    3960
[Chinese]          2735
[Fast, Food]       1986
[Mughlai]           995
[Italian]           764
[Bakery]            745
[Continental]       736
[Cafe]              703
[Desserts]          653
[South, Indian]     636
Name: count, dtype: int64
```



Cost & Rating Analysis:

- Analysed cost distribution using histograms

- Studied rating distribution

- Compared ratings across cost bands, delivery and booking options

Please find detailed code and charts in MarketingCapstone.ipynb file.

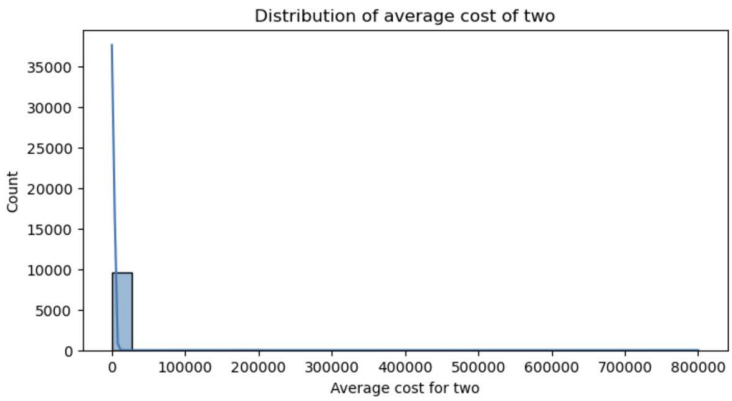
Distribution of Cost Across Restaurants

```
data_merged['Average Cost for two'].describe()
```

```
count      9551.000000
mean       1199.210763
std       16121.183073
min           0.000000
25%         250.000000
50%         400.000000
75%         700.000000
max      800000.000000
Name: Average Cost for two, dtype: float64
```

```
# Cost distribution plot

plt.figure(figsize=(8,4))
sns.histplot(data_merged['Average Cost for two'],bins=30,kde=True)
plt.title('Distribution of average cost of two')
plt.xlabel('Average cost for two')
plt.show()
```



6. Tableau: Data Connection & Modelling

Step 1: Import Data into Tableau

- Connected cleaned dataset from Python output

- Verified schema and field roles

Step 2: Relationships

- Ensured Country information correctly mapped

- Validated joins/relationships

Step 3: Data Role Adjustments

- Set geographical roles for Country and City

- Converted measures and dimensions appropriately

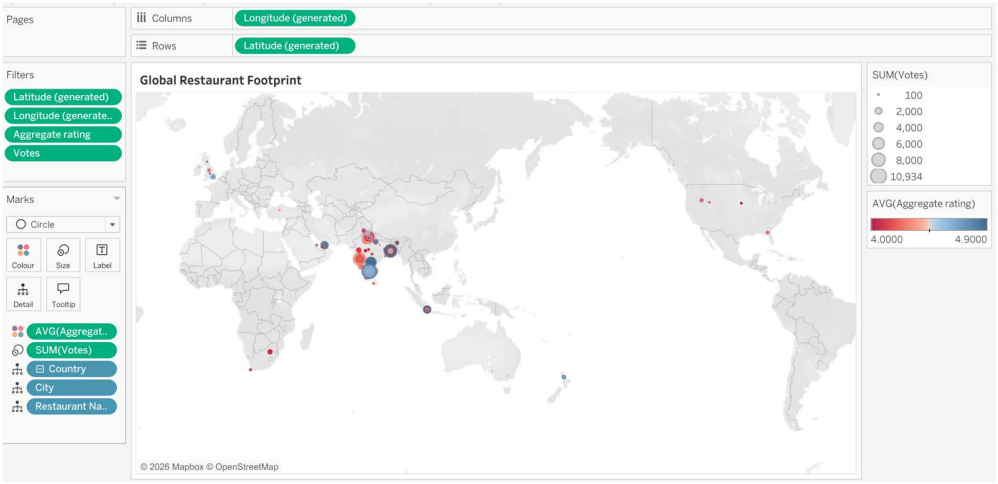
7. Tableau: Worksheet Development

Created the following visualisations:

- Map: Restaurant distribution by Country and City

  - Size: Votes

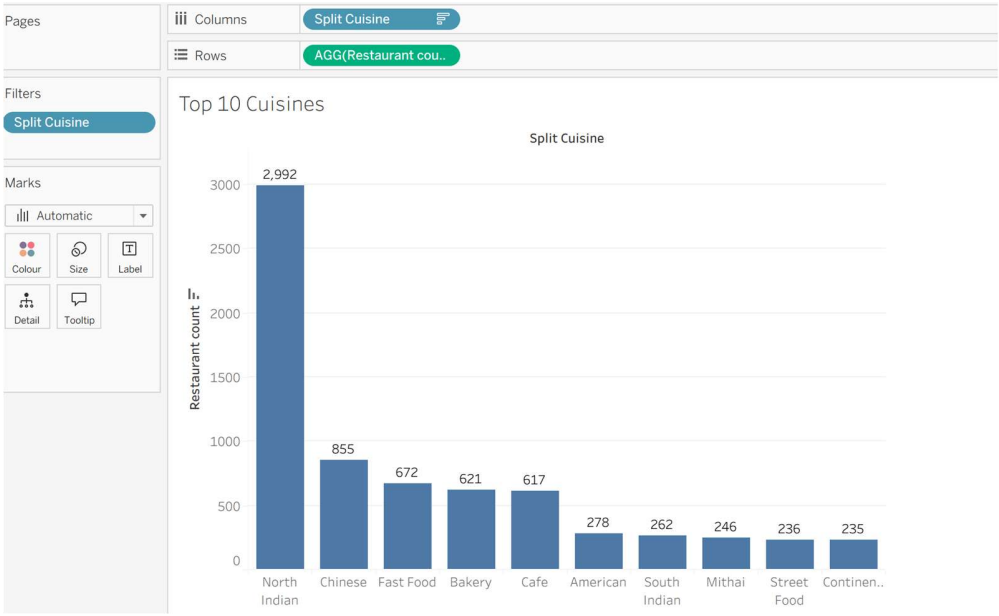
  - Colour: Average Rating



- Bar Charts:

  - Top Cities by Restaurant Count

  - Top 10 Cuisines



- KPI Cards:

  - Total Restaurants

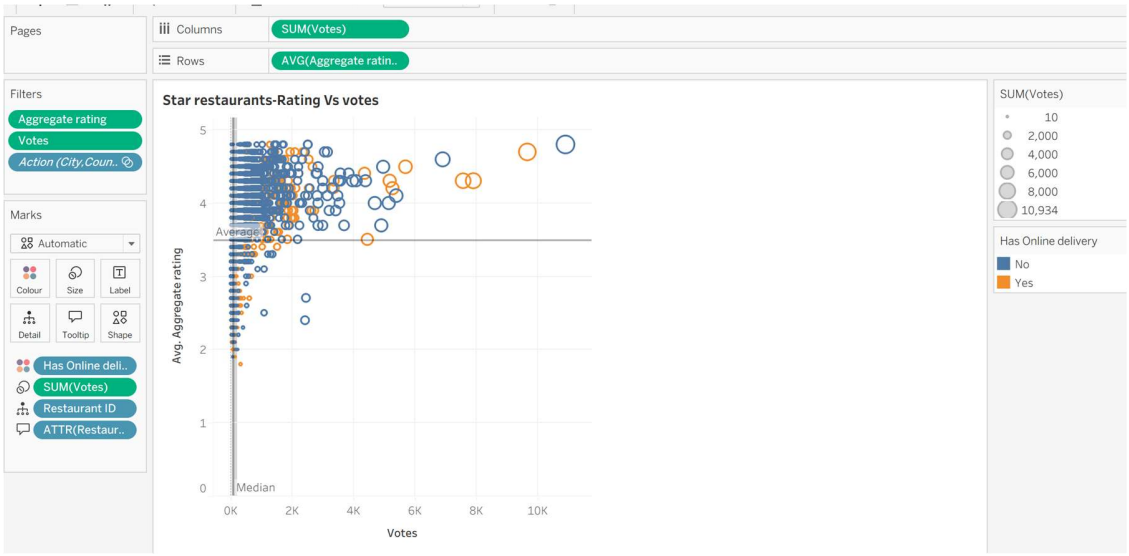
  - Average Rating

  - % Online Delivery

- % Table Booking

- Scatter Plot:

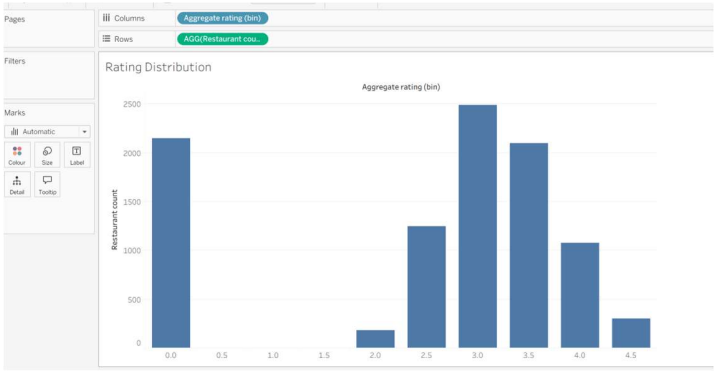
  - Votes vs Ratings to highlight star restaurants

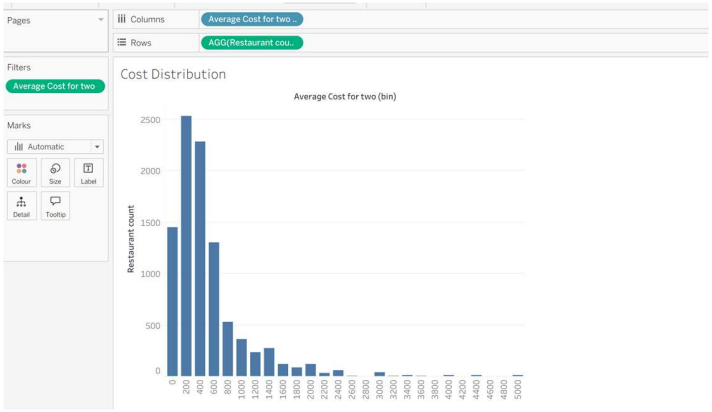


- Distribution Charts:

  - Cost distribution

  - Rating distribution





8. Tableau: Dashboard Design

Dashboard Objectives:

- Identify star restaurants quickly

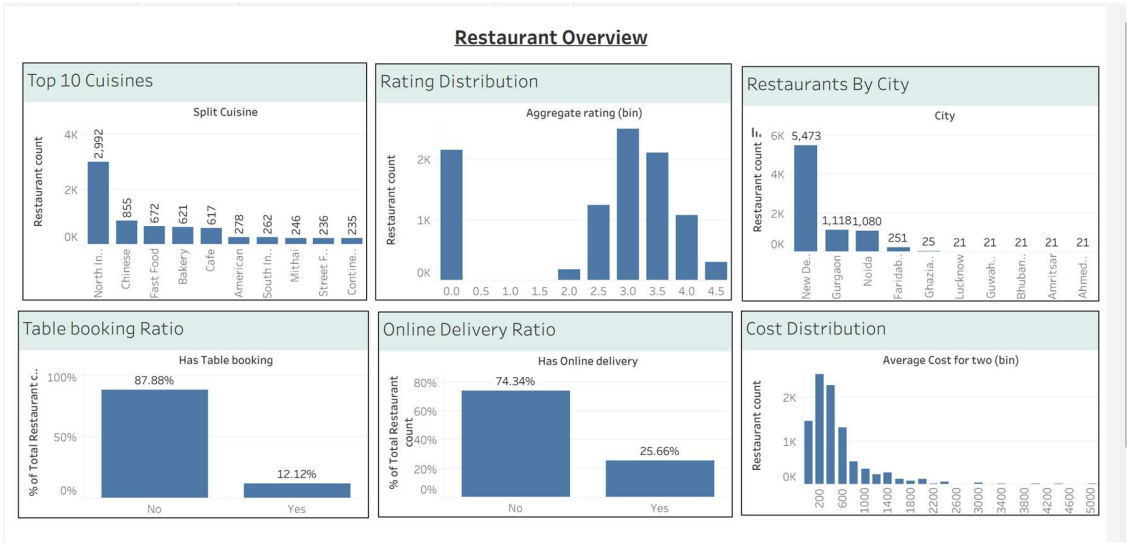
- Explore geographical and behavioural patterns

Dashboard Features:

- Interactive filters: Country, City, Delivery, Booking, Cost, Rating

- Cross-filtering and highlight actions

- Clean layout with logical grouping of KPIs, maps, and charts



9. Star Restaurant Identification Logic

Star restaurants were identified using a multi-dimensional approach:

- High average ratings
- Strong vote counts (popularity)
- Online delivery availability
- Table booking availability
- Balanced cuisine variety
- Contextual cost considerations

This approach avoids bias from ratings alone and improves recommendation quality.

10. Outcome & Business Value

Key Outcomes:

- Delivered a clean analytical dataset
- Identified performance drivers for restaurants
- Built an interactive Tableau dashboard
- Enabled data-backed recommendations for B2C portal enhancement

Business Value:

- Improved restaurant discovery
- Better customer experience
- Scalable foundation for intelligent automation